

Protocol Layering

Protocol Layering

- ▶ In computer networks, communication between devices follows a set of rules called a **protocol**.

A protocol defines:

- ▶ How data is transmitted
- ▶ How devices communicate
- ▶ How errors are handled
- ▶ How information is interpreted
- ▶ When communication becomes complex, a single protocol is not enough. The communication task is therefore divided into multiple smaller tasks organized into different layers. This concept is called:

Protocol Layering

Protocol Layering

Definition

- ▶ *Protocol layering is a method of dividing a communication task into several smaller and simpler layers, where each layer performs a specific function and follows its own protocol.*

Why Protocol Layering is Needed

Modern communication systems are complex because they involve:

- ▶ Data transmission
- ▶ Error handling
- ▶ Encryption
- ▶ Routing
- ▶ Delivery mechanisms
- ▶ Instead of handling all tasks together, networking divides the work into layers.

Why Protocol Layering is Needed

Each layer:

- ▶ Performs a specific task
- ▶ Provides services to the upper layer
- ▶ Receives services from the lower layer

First Scenario: Single-Layer Communication

In this :

- ▶ Maria and Ann communicate face-to-face
- ▶ Communication happens directly in one layer

Figure 1.9 *A single-layer protocol*



First Scenario: Single-Layer Communication

Even in simple communication, certain rules are followed:

- ▶ Greeting each other
- ▶ Speaking one at a time
- ▶ Using understandable language
- ▶ Taking turns in conversation

These rules form a simple protocol.

Second Scenario: Three-Layer Communication

- ▶ When Ann moves to another city, communication becomes more complex.
- ▶ Maria and Ann now communicate through postal mail and use encryption for security.

Second Scenario: Three-Layer Communication

Their communication is divided into three layers.

- ▶ Maria first creates a message in plain English at the third layer. The message is then passed to the second layer, where it is encrypted into ciphertext. Next, the first layer places the encrypted message in an envelope, adds the addresses, and sends it through the postal system.
- ▶ At Ann's side, the first layer receives the mail and extracts the ciphertext from the envelope. The second layer decrypts the ciphertext and converts it back into plaintext. Finally, the third layer reads the message as if Maria were speaking directly to Ann.

Second Scenario: Three-Layer Communication

This example shows how protocol layering divides communication into smaller tasks such as:

- ▶ Message creation
- ▶ Encryption/decryption
- ▶ Sending and receiving mail

Each layer performs a specific function independently.

Three Layers in the Example

Layer	Function
Layer 3	Listen and Talk
Layer 2	Encrypt and Decrypt
Layer 1	Send and Receive Mail

Working of the Three-Layer Protocol

At Maria's Side

▶ Layer 3

- ▶ Maria creates the message (plaintext)

▶ Layer 2

- ▶ Message is encrypted into ciphertext

▶ Layer 1

- ▶ Ciphertext is placed in an envelope and mailed

Working of the Three-Layer Protocol

At Ann's Side

▶ Layer 1

- ▶ Mail is received

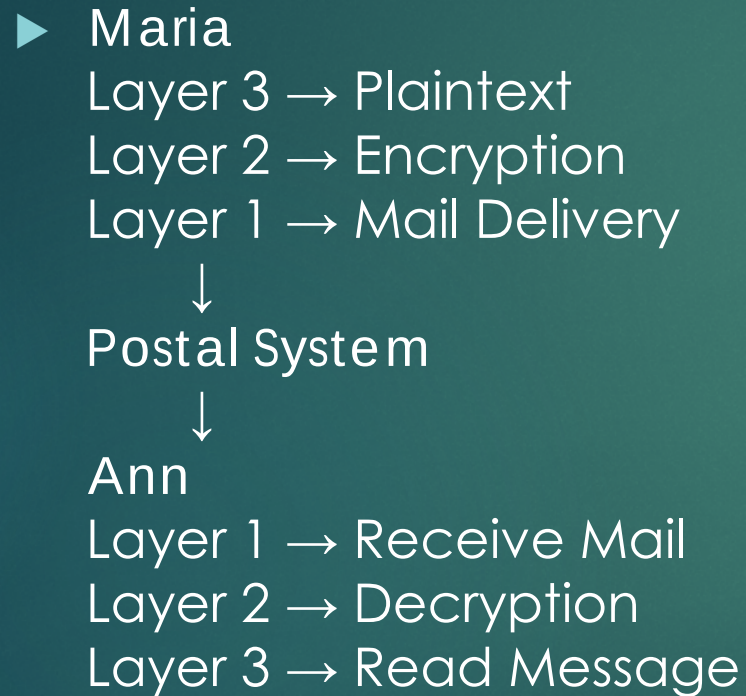
▶ Layer 2

- ▶ Ciphertext is decrypted into plaintext

▶ Layer 3

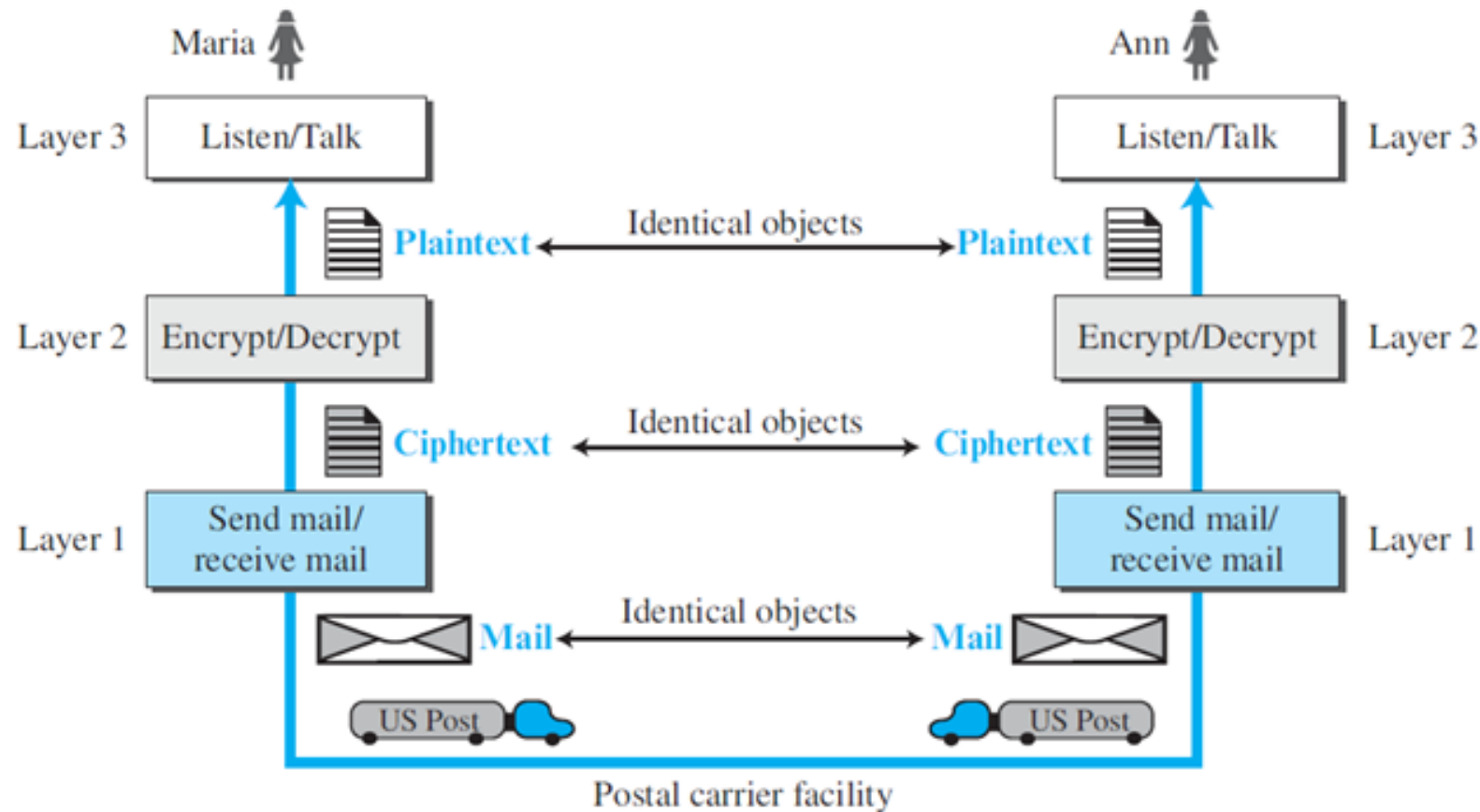
- ▶ Ann reads the message

Simple Representation

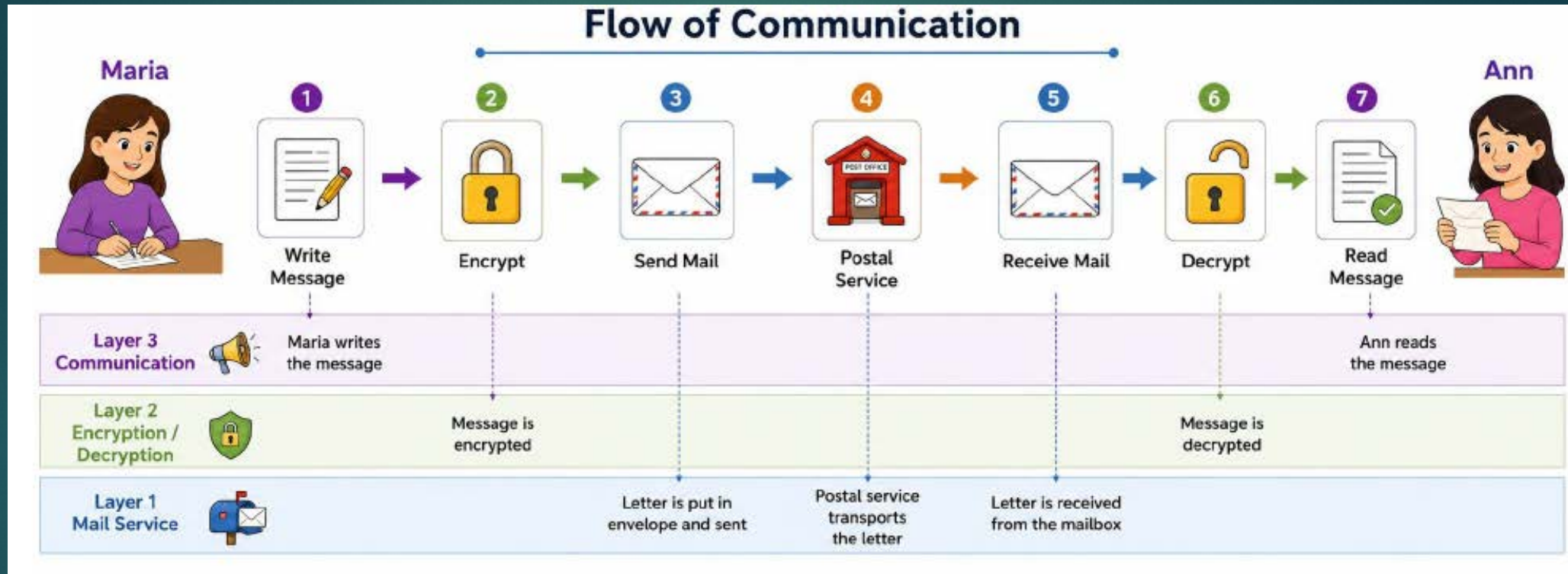


Simple Representation

Figure 1.10 A three-layer protocol



Simple Representation



Advantages of Protocol Layering

1. Modularity

- ▶ Each layer works independently.
- ▶ If one layer changes:
- ▶ Other layers remain unaffected

Example:

- ▶ Maria and Ann can replace only the encryption layer without changing the mailing system.

Advantages of Protocol Layering

2. Simplicity

- ▶ Complex communication is divided into smaller manageable tasks.
- ▶ Each layer focuses on one responsibility.

Advantages of Protocol Layering

3. Flexibility

- ▶ Different implementations can be used as long as:
- ▶ Input and output remain the same

Example:

- ▶ Encryption machines from different manufacturers can still work correctly.

Advantages of Protocol Layering

4. Easy Maintenance

- ▶ Problems can be isolated to a specific layer.
- ▶ This simplifies:
 - ▶ Troubleshooting
 - ▶ Upgrading
 - ▶ Testing

Advantages of Protocol Layering

5. Cost Reduction

- ▶ Intermediate devices do not need all layers.

Example:

- ▶ Routers need only some networking layers
- ▶ End systems need all layers
- ▶ This reduces complexity and cost.

Principles of Protocol Layering

1. Bidirectional Communication

- ▶ Each layer must support communication in both directions.

Examples:

- ▶ Listen / Talk
- ▶ Encrypt / Decrypt
- ▶ Send / Receive

Principles of Protocol Layering

2. Identical Objects at Peer Layers

- ▶ Objects exchanged between corresponding layers should be identical.

Examples:

- ▶ Layer 3 → Plaintext
- ▶ Layer 2 → Ciphertext
- ▶ Layer 1 → Mail

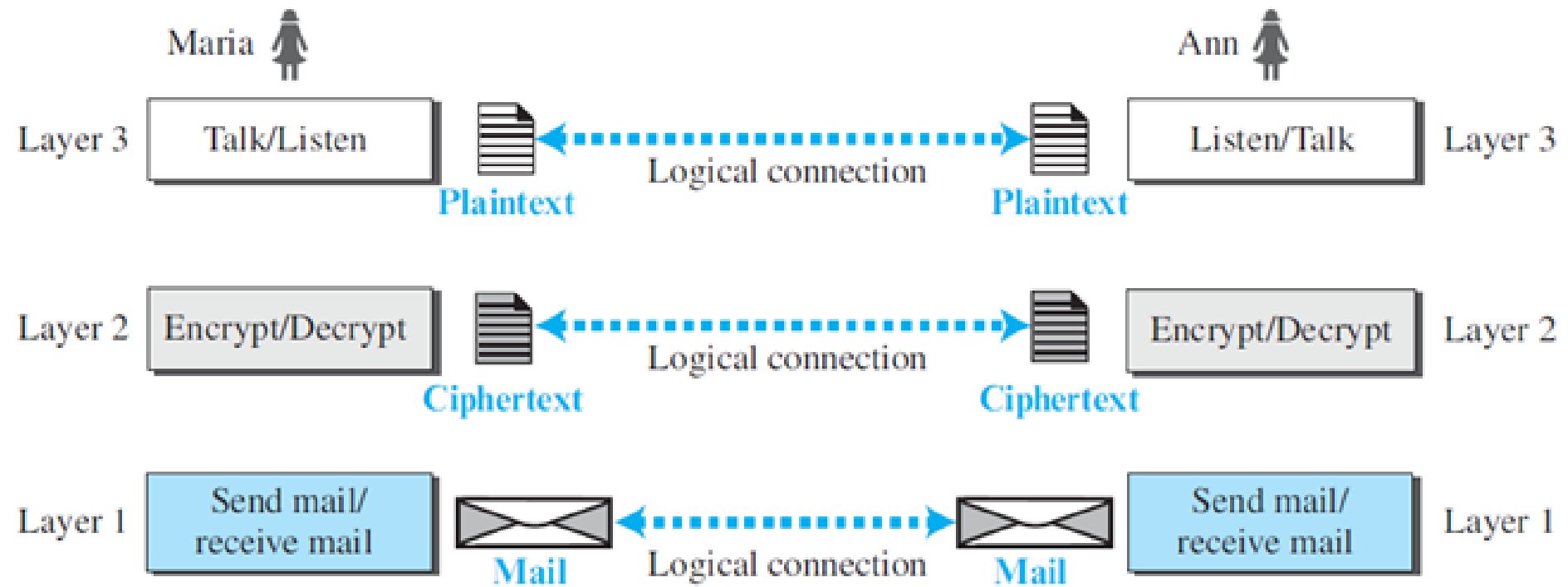
Logical Connections

In protocol layering:

- ▶ Each layer appears to communicate directly with its corresponding layer at the other side.
- ▶ This is called:
- ▶ ***Logical Layer-to-Layer Communication***
- ▶ Although actual data travels through lower layers, logically each layer communicates with its peer layer.

Logical Connections

Figure 1.11 *Logical connection between peer layers*



Importance in Computer Networks



Protocol layering forms the basis of:

- ▶ Internet communication
- ▶ OSI model
- ▶ TCP/IP model

It allows modern networks to:

- ▶ Scale efficiently
- ▶ Support multiple technologies
- ▶ Simplify communication design

Real-Life Analogy

Sending a parcel through courier service:

Each layer performs a separate task independently.

Layer	Function
Writing Layer	Create message
Packaging Layer	Wrap and secure
Delivery Layer	Transport parcel

Summary

- ▶ *Protocol layering is the process of dividing complex communication into multiple smaller layers, where each layer performs a specific task and follows its own protocol. This approach provides modularity, flexibility, simplicity, and efficient communication, forming the foundation of modern computer networking systems.*

TCP/IP Protocol Suite

- ▶ The **TCP/IP Protocol Suite** is the set of communication protocols used in the Internet today.
- ▶ TCP/IP stands for:
- ▶ **Transmission Control Protocol (TCP)**
- ▶ **Internet Protocol (IP)**
- ▶ It is a collection of protocols organized into layers, where each layer performs a specific task in communication.
- ▶ *The TCP/IP protocol suite is a hierarchical set of protocols organized into layers that enables communication over the Internet.*

Why TCP/IP Uses Layering

- ▶ Communication in networks is complex.

TCP/IP divides communication into smaller layers so that:

- ▶ Each layer performs a specific function
- ▶ Layers work independently
- ▶ Communication becomes easier to manage
- ▶ Each upper layer receives services from the lower layer.

Why TCP/IP Uses Layering

- ▶ **Layer Interaction**

- ▶ Each layer provides services to the layer above it and receives services from the layer below it. This structured approach ensures efficient, reliable, and organized communication across networks.

- ▶ **In simple terms, the TCP/IP model divides a complex communication process into smaller, manageable layers, making Internet communication efficient and reliable.**

TCP/IP Layered Architecture

- ▶ Originally, TCP/IP had four layers.
Today, it is commonly represented as a **five-layer model**.

Five Layers of TCP/IP

Layer	Function
Application Layer	Provides services to users and applications
Transport Layer	End-to-end communication and reliability
Network Layer	Routing and delivery of packets
Data Link Layer	Node-to-node communication
Physical Layer	Transmission of bits through media

TCP/IP Five-Layer Model

Figure 1.12 *Layers in the TCP/IP protocol suite*

